

TRUSTIA AI — NATO DIANA ACCELERATOR DOSSIER

DEFENCE INNOVATION PROPOSAL // DUAL-USE AUTONOMY IN GPS-DENIED OPERATING THEATRES (€400,000 GRANT)

GRANT VALUE: €400,000 GRANT 0% Equity / Non-Dilutive	INTEROPERABILITY: STANAG 4586 NATO Level 4 / JAUS	CODE & TESTS: 1,281 TESTS (100%) 16K Lines Production	DEFENSE RATING: SSB 100/100 ITO BTM & Teknopark
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1. Executive Summary & Operational Defense Problem

TRUSTIA AI is a deterministic, hardware-agnostic Level 4 autonomy software stack engineered specifically for tactical Unmanned Ground Vehicles (UGVs) and dual-use mobility platforms operating in intense electronic warfare and GPS-denied combat environments across NATO Allied theaters.

The Threat: Electronic warfare systems completely sever satellite GNSS positioning in modern peer conflicts. Commercial stacks fail catastrophically under jamming. Furthermore, defense primes spend €10M–€25M and 3–5 years rewriting autonomy software for every mechanical vehicle chassis. Trustia delivers a plug-and-play universal software "brain" converting any vehicle into a Level 4 system in under 15 minutes with sub-8cm drift across 10 km.

2. Technological Innovation & NATO Interoperability

400Hz GPS-Denied ESKF	3D LiDAR + 400Hz IMU sensor fusion. Position drift <0.08m over 10km in electronic warfare theaters.	Anti-Jamming / Anti-Spoofing
Universal CAN Integration	Direct CAN 2.0B / CAN FD / SAE J1939 Drive-by-Wire interface. Vehicle conversion in < 15 minutes.	SAE J1939 / ISO 26262
1,281 Automated Tests	Zero external critical dependencies. Deterministic real-time cycle execution latency < 1.2 ms.	1,281 / 1,281 (100% PASS)
Multimodal Threat Isolation	LWIR Thermal (FLIR Boson) + Induction + GPR Radar fusion to isolate explosive hazards autonomously.	NATO STANAG 4586 Level 4

3. Dual-Use Operational Applications & Allied Market Scalability

- Allied Defence Operations (B2G): Autonomous convoy resupply in hostile corridors (eliminating driver casualties), autonomous route clearance, landmine detection perimeters, and perimeter security.
- Commercial Applications (B2B): Heavy autonomous mining trucks (CAT/Komatsu), agricultural platforms, and urban robotaxis.
- TAM / SAM: \$34.8B military UGV market + \$120B commercial autonomous vehicle market by 2032.

Projected Total Revenue (€)	€250.000	€1.200.000	€4.100.000
Integrated Vehicle Platforms	6 Vehicles (PoC)	35 Platforms	150+ Platforms (Allies)
Annual C2 / Tactical SaaS Licenses	€90.000	€550.000	€2.400.000
Software Gross Margin (%)	84%	87%	91%
Engineering Team Size	4 R&D; Engineers	10 Professionals	22 Professionals

4. Founder Qualifications & Sovereign Track Record

- Founder & Chief Executive Officer (CEO): Murat Furkan Bayram.
- Institutional Backing: Admitted to Istanbul Chamber of Commerce BTM Incubator (Polat Tower, Istanbul).
- Defense Tech Hub: Candidate defense technology company in Teknopark Istanbul (Cube Incubation / Cube GO).
- Defense Industry Excellence Rating: 100/100 Perfect Grade by Turkish Presidency of Defense Industries (SSB & BTK, ID: L2zPtN4X1ZJ).
- Certified Advanced Entrepreneur: Accredited by the Ministry of Industry and Technology (KOSGEB KSB01UGE0115153370).
- National R&D; Researcher: Registered in TÜBİTAK ARBİS (TBTk-0229-6571). Active ASELSAN Defense Supplier Portal Registration.

5. NATO DIANA Next Steps & Implementation

PHASE 1 (€100,000 GRANT): Accelerate live tactical vehicle integration trials at NATO-affiliated test centers.
PHASE 2 (€300,000 GRANT): Deploy interoperable NATO STANAG 4586 Level 4 autonomy across Allied defense partner fleets.
SUBMISSION: Direct portal entry at diana.nato.int under the Autonomy & Resilient Sensing Challenge.